



ON NONPARAMETRIC ESTIMATION OF A HEDONIC PRICE FUNCTION

HARRY HAUPT,^{a*} JOACHIM SCHNURBUS^b AND ROLF TSCHERNIG^b

^a *Department of Business Administration and Economics, Bielefeld University, Centre for Statistics, Department of Bielefeld, Germany*

^b *University of Regensburg, Regensburg, Germany*

SUMMARY

Recently, using mixed data on Canadian housing, Parmeter, Henderson, and Kumbhakar (*Journal of Applied Econometrics* 2007; **22**: 695–699) found that a nonparametric approach for estimating a hedonic house price function is superior to formerly suggested parametric and semiparametric specifications. We carefully reanalyze these specifications for this dataset by applying a recent nonparametric specification test and simulation-based prediction comparisons. For the case at issue our results suggest that a previously proposed parametric specification does not have to be rejected and we illustrate how nonparametric methods provide valuable insights during all modeling steps. Copyright © 2010 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Hedonic theory suggests that regressions concerning the heterogeneous good housing should reflect the intrinsic nonlinearity in the relationship between house prices and housing characteristics, though nothing is known a priori about a specific functional form. Consequently, regression methods should impose as little parametric structure as possible. Therefore, semi- and nonparametric regressions suggest themselves as ideal tools for modeling hedonic prices. Anglin and Gencay (1996), hereafter AG, confirm this presumption. On the basis of a dataset on the Canadian housing market and various specification tests, they find semiparametric methods to be superior to a parametric specification motivated by Box–Cox transformations. Using the same data, Parmeter *et al.* (2007), hereafter PHK, find a nonparametric approach superior to the specifications considered by AG where they apply specifically the nonparametric kernel regression method suggested by Li and Racine (2004, 2007) and Racine and Li (2004) that allows appropriate smoothing of both continuous and discrete regressors. Comparing their parametric and nonparametric specifications,

* Correspondence to: Professor Harry Haupt, Centre for Statistics, Department of Business Administration and Economics, Bielefeld University, PO Box 100131, 33501 Bielefeld, Germany. E-mail: hhaupt@wiwi.uni-bielefeld.de

PHK observe an in-sample fit measured by an appropriate pseudo R^2 that is about 10% larger for the latter.

In this paper we reconsider the Canadian housing dataset and several of the models used by AG and PHK with respect to general specification testing and variable selection, but also with respect to out-of-sample prediction performance. We apply the misspecification test of Hsiao *et al.* (2007), which explicitly allows for mixed continuous and discrete regressors and has not yet been applied to this dataset. Additionally, we conduct simulation-based prediction comparisons and tests. In total, we consider three model specifications: (i) a parametric log-log model considered by AG (see Table III in AG). Due to taking logs of the dependent price variable it constitutes the simplest model allowing for nonlinear marginal effects with respect to housing prices and is our benchmark model; (ii) the nonparametric model applied by PHK; (iii) a slightly more general version of the semiparametric model of AG. In Section 2 we present evidence that the parametric benchmark model performs better than both the semi- and nonparametric regression model. In Section 3 we illustrate how a careful analysis of estimated bandwidths and applied kernels allows additional insights into the relevance of regressors and/or their functional impact. Section 4 concludes.

2. SPECIFICATION TESTING AND PREDICTION PERFORMANCE

AG and PHK analyze a dataset for the Canadian housing market, where the conditional mean of the natural log of price is explained by 11 characteristics of a house. The dataset consists of 546 observations, each with one continuous regressor, the lot size (in logs), and 10 discrete regressors, six of them binary. AG apply different parametric specifications, as well as the semiparametric estimator of Robinson (1988). PHK reject the null hypothesis of correct semiparametric specification using the nonparametric test of Delgado and González Manteiga (2001). To allow for additional flexibility PHK apply the nonparametric generalized kernel estimation method presented in detail by Li and Racine (2004, 2007) and Racine and Li (2004). In our study we begin from scratch and consider the parametric model of AG (Table III) as our null model and use the specification test of Hsiao *et al.* (2007). This test is based on nonparametric generalized kernel estimation and is—for a setting of mixed regressors—better suited than traditional nonparametric tests.¹ Thus a rejection of the null hypothesis would provide further evidence against the parametric model.²

2.1. Nonparametric Test of Hsiao, Li, and Racine

Computing variants of the Hsiao *et al.* (2007) test require several specification choices. One has to choose a kernel function for each type of variable (continuous, ordered/unordered categorical), a bandwidth selector, and the degree of the local polynomial estimator. Moreover, p -values can be based on the asymptotic distribution as well as on bootstrap distributions. Table I exhibits the p -values for several different specifications. Results in the left panel of Table I are based on the kernel functions of Aitchison and Aitken (1976) and of Wang and van Ryzin (1981) for

¹ In traditional tests, the sample is split up for the category combinations of the discrete regressors.

² We do not test the semiparametric specification versus the nonparametric specification as PHK do since the Hsiao *et al.* (2007) test requires a parametric null model.

Table I. Bootstrap p -values for various specifications of the test of Hsiao *et al.* (2007)

	AA-WvR				LR-LR			
	LSCV		AIC _c		LSCV		AIC _c	
	LC	LL	LC	LL	LC	LL	LC	LL
<i>i.i.d. bootstrap</i>								
lnlot	0.1278	0.1805	0.0802	0.0226	0.1053	0.1128	0.0877	0.0526
lot	0.1955	0.1454	0.0702	0.0251	0.1805	0.1504	0.0877	0.0802
<i>Wild bootstrap</i>								
lnlot	0.1529	0.2180	0.1103	0.0351	0.1454	0.1429	0.1328	0.0727
lot	0.2607	0.0927	0.1128	0.0401	0.2607	0.2055	0.1579	0.1128

unordered and ordered discrete regressors (AA-WvR), respectively. Results in the right panel are produced with the corresponding kernels of Li and Racine (LR-LR) (for details, see Racine and Li, 2004). Throughout, a Gaussian kernel is used for the continuous regressor. Furthermore, the configurations differ in the degree of local polynomial estimation (local constant (LC) and local linear (LL)) and the bandwidth selection method (least-squares cross-validation (LSCV) and the AIC-based method of Hurvich *et al.* (1998) (AIC_c)).³ The line entries *lnlot* and *lot* indicate whether the underlying continuous regressor enters the nonparametric estimation within the Hsiao *et al.* (2007) test in logs or in levels. We used both configurations since it is not obvious whether to transform a continuous regressor prior to nonparametric estimation. We do not exhibit p -values based on the asymptotic distribution as they are all larger than 0.27, while all bootstrap p -values are lower than 0.27. This is in line with improving power by bootstrapping asymptotically pivotal tests.

Several conclusions can be drawn from the estimated p -values in Table I. First, by comparing the entries for *lnlot* and *lot* we can see that the p -values fluctuate to some extent—however, without generally changing rejection decisions at typical significance levels. So, for the test it does not matter much whether the continuous regressor is transformed or not. Second, we find the choice of bandwidth selection method to have a substantial impact. For both bootstrap methods, p -values based on AIC_c are smaller than those based on LSCV. This drop is especially pronounced for the local linear estimator (LL), particularly for the Aitchison and Aitken (1976) and Wang and van Ryzin (1981) kernel (AA-WvR). Thus the choice of bandwidth selection method matters a lot. Third, the choice of kernels in general has a rather small impact. If p -values are around a chosen significance level, however, the test decision can be influenced by the kernel choice, as occurs for the AIC_c, LL setup at a significance level of 0.05.

In sum, for the given dataset the Hsiao *et al.* test (2007) is sensitive with respect to its specification.⁴ The only test specifications leading to a rejection of the null hypothesis at a significance level of 0.05 are local linear estimates with the AA-WvR kernels in combination

³ Note that all of our calculations are done in R (version 2.9.1) and the nonparametric estimations and tests are conducted with the np-package (version 0.30–3 of Hayfield and Racine, 2009). To ensure reproducibility, we use the seed 42 for all calculations in the paper where it is appropriate and the default settings for the functions of the np-package unless otherwise stated. All the results are calculated using 20 multistarts for the bandwidth selections.

⁴ Note that the specification (LR-LR, LSCV, LC) suggested by Hsiao *et al.* (2007) leads to p -values of about 0.14 and 0.26, respectively.

with AIC_c bandwidth selection. For all other specifications, the estimated p -values are larger than 0.05.

2.2. Monte Carlo Prediction Simulation

Since various specifications of the Hsiao *et al.* test (2007) do not provide a clear-cut answer, we need another tool for evaluating the parametric benchmark model. We choose to compare the predictive abilities of the parametric benchmark specification and the nonparametric specification of PHK. This is accomplished by means of Monte Carlo simulation techniques. For completeness, we also include a semiparametric specification. It is based on equation (11) in AG but, in contrast to AG, using the kernel of Wang and van Ryzin (1981) for the ordered discrete regressors in the nonparametric part and allowing for a different bandwidth for each regressor.⁵ For each replication we randomly split the sample into two subsamples: 90% of the observations are taken for computing the estimates that in turn are used to predict (log-)prices for the remaining 10% of the houses. These predictions are used to compute the average squared error of prediction (ASEP) for each replication.⁶ We apply the nonparametric specification that exhibits the smallest p -value in each row of Table I (AA-WvR, AIC_c , LL) and conduct 10,000 replications.

Figure 1(a) shows the distribution functions resulting from the Monte Carlo simulation over the whole range of simulated ASEPs. Vertical lines to the left and right of each distribution function indicate the smallest and largest ASEP obtained for each model specification. Results for the parametric, semiparametric, and nonparametric specification are plotted in solid, dashed, and dotted lines, respectively.

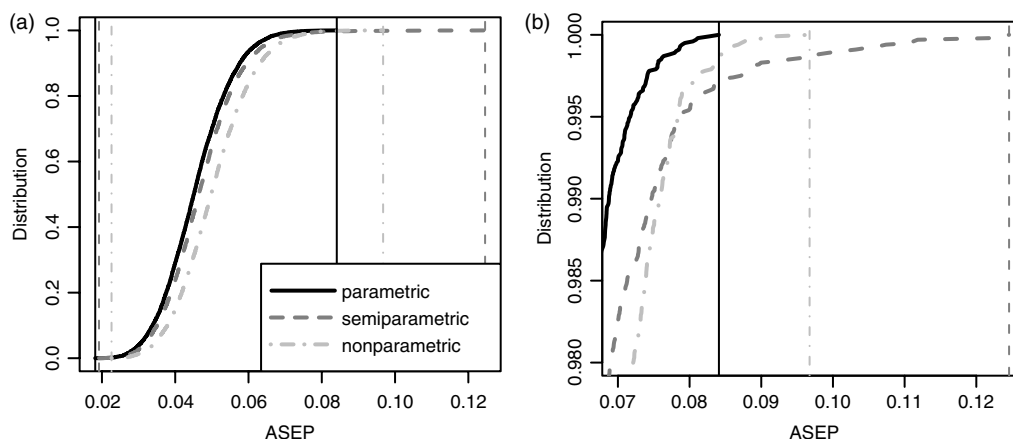


Figure 1. Simulation-based distributions of the ASEP. Vertical lines indicate the lowest and highest value of ASEP for each specification: (a) full range; (b) upper tail

⁵ An alternative is the setup of AG, who treated the discrete regressors in the nonparametric part of the partial linear model as continuous variables and used the same bandwidth for all regressors and in all nonparametric regressions in the first step of Robinson's (1988) method. We would like to mention that AG and others also investigated the out-of-sample prediction performance, but without random sampling and with one replication only.

⁶ For one replication the ASEP of the n_p observations in the prediction sample is given by $\sum_{i=1}^{n_p} (\hat{y}_i - y_i)^2 / n_p$, where y_i and $\hat{y}_i$ denote the actual observation and its prediction, respectively.

dashed, and dashed-dotted lines, respectively. Figure 1(b) provides a zoom on the crossing of the distribution functions in the upper tail. Comparing the Monte Carlo distribution functions in Figure 1(a) and (b) shows that the distribution function of the parametric specification is always to the left of the other two. Thus the parametric benchmark specification first-order stochastically dominates (cf. Section 4.3.6 in Vinod, 2008) both the semi- and nonparametric specification. This finding is reflected in the frequencies of how often the parametric specification predicts better than one of its competitors, where the parametric specification predicts better than the semi- and nonparametric specification in 66% and 83% of all replications, respectively.

Finally, testing whether the means of the ASEPs significantly differ between each specification confirms the previous results. The null hypothesis that the mean of the ASEP for the parametric specification is larger or equal than that for the nonparametric specification is rejected. The parametric specification also has a significantly lower mean ASEP than the semiparametric specification. The corresponding p -values are all less than 0.0001.⁷

In sum, the parametric benchmark model predicts best. In detail, a visual comparison of the distribution functions of the semi- and nonparametric specification displayed in Figure 1(a) suggests that the semiparametric specification has lower ASEPs. It is evident from Figure 1(b) that the distribution functions cross and the semiparametric specification has by far the largest ASEP values across all specifications. Thus the semiparametric specification sometimes fails drastically and has the largest probability of having a very bad prediction performance. Nevertheless it predicts better than the nonparametric specification in 72% of all replications and exhibits a significantly lower mean ASEP.⁸ All the previous results remain qualitatively valid for alternative sample-split proportions.

3. COMMENTS ON THE NONPARAMETRIC PERFORMANCE

A nonparametric model is expected to represent the data-generating mechanism better than a competing parametric model if the latter misses important features of the functional shape and the data provide enough information for the nonparametric estimation. In this vein a detailed analysis of estimated bandwidths and applied kernels allows additional insights into the relevance of regressors (cf. Section 4.5 in Li and Racine, 2007) and/or their functional impact.

Using again the AA-WvR, AIC_c, LL setup, Table II displays the estimated bandwidths, types of regressors, applied kernels and maxima of admissible bandwidth ranges. We note that this setup reproduces the gradient values given in Table I of PHK. The first row displays the results for the only continuous regressor, `lnlot`. The estimated bandwidth is extremely large relative to the observed values of the regressor. This strongly suggests that no smoothing is needed for this variable in the generalized product kernel. For unordered discrete regressors the Aitchison and Aitken (1976) weighting function allows irrelevant regressors to be smoothed out due to identical weights across all categories of the corresponding regressor. This occurs for binary regressors if the bandwidth equals its upper bound 0.5. Table II reveals that the variable `rec` (recreational room exists/does not exist) is irrelevant for the estimated house price since the estimated bandwidth is in the close vicinity of 0.5.

⁷ Given the very small p -value we do not correct for the dependence among replications in paired t -tests.

⁸ At first sight this seems to contradict the specification test of Delgado and González Manteiga (2001) applied by PHK. As already noted, however, PHK use AG's more restricted semiparametric model.

Table II. Estimated bandwidths of local linear kernel regression

Regressor	Regressor type	Weighting function	Est. bandwidth	Max. bandwidth
lnlot	Continuous	Gaussian (2. order)	1,021,545	∞
ca	Binary	Aitchison and Aitken	0.1409	0.5
drv	Binary	Aitchison and Aitken	0.1206	0.5
ffin	Binary	Aitchison and Aitken	0.1296	0.5
ghw	Binary	Aitchison and Aitken	0.0302	0.5
rec	Binary	Aitchison and Aitken	≈ 0.5	0.5
reg	Binary	Aitchison and Aitken	0.2253	0.5
bdms	Ordered	Wang and van Ryzin	0.6153	< 1
fb	Ordered	Wang and van Ryzin	0.2667	< 1
gar	Ordered	Wang and van Ryzin	≈ 1	< 1
sty	Ordered	Wang and van Ryzin	0.5063	< 1

The remaining discrete regressors are treated as ordered with the weighting function of Wang and van Ryzin (1981). This kernel does not allow for smoothing out irrelevant regressors, which may lead to inefficient estimation. This limitation becomes apparent by inspecting the Wang and van Ryzin kernel function:

$$l(\delta, \lambda) = \begin{cases} 1 - \lambda & \text{for } \delta = 0, \\ \frac{1}{2}(1 - \lambda)\lambda^{|\delta|} & \text{for } \delta \in \mathbb{Z} \end{cases}$$

where λ is the selected bandwidth for the discrete regressor and δ represents the distance between the category of interest and any category used for estimation. Note that a bandwidth λ of exactly 1 implies zero kernel weights for any δ and thus is not admissible. In contrast, a zero bandwidth implies that only those observations are used that are in the category of interest, implying no smoothing at all. Thus, for smoothing to occur, the bandwidth has to take values in $0 < \lambda < 1$. Now consider the ratio of the kernel weights for the category of interest, $\delta = 0$, and any other category $\delta \neq 0$. As an example, you may be interested in the price for a house that has two garages ($\text{gar} = 2$), so all observations with two garages exhibit $\delta = 0$ and are therefore weighted by $1 - \lambda$, while observations with one or three garages have $|\delta| = 1$ and are assigned weights $\frac{1}{2}(1 - \lambda)\lambda$, respectively. In general, the ratio of weights equals

$$\frac{1 - \lambda}{\frac{1}{2}(1 - \lambda)\lambda^{|\delta|}} = \frac{2}{\lambda^{|\delta|}}, \quad 0 < \lambda < 1$$

where equality only holds within the indicated range of bandwidths. Clearly, this ratio decreases monotonically with an increasing bandwidth and approaches its infimum 2 as the bandwidth λ tends to 1. Thus this ratio cannot take any value smaller than 2. This clearly limits the amount of smoothing since (almost) equal weights are not possible. In particular, equal weights corresponding to a ratio of 1 are ruled out and therefore smoothing out of irrelevant variables cannot occur. Inspecting the remaining four rows of Table II shows that this limitation becomes relevant for the ordered regressor gar . The corresponding bandwidth is estimated to be extremely close to 1, with a precise value of 0.99999999999944. This indicates that more smoothing might be needed. Complete smoothing is allowed for by the kernel of Li and Racine. Using the latter the bandwidth estimate for gar is 0.6876. Choosing $|\delta| = 1$ the corresponding ratio of weights is $\lambda^{-|\delta|} = 1.45$,

indicating the relevance of this variable on the one hand and the need for strong smoothing on the other hand.

4. CONCLUSION

In this replication study of the work of Parmeter *et al.* (2007), who applied and advocated nonparametric techniques for estimating hedonic house price functions, we re-estimate their nonparametric specification and compare it to parametric and semiparametric specifications formerly proposed by Anglin and Gencay (1996). We extend their analysis by using a recently suggested nonparametric specification test for mixed continuous and categorical data (Hsiao *et al.*, 2007) and simulation-based prediction comparisons. It turns out that test results depend on the choice of kernels, of the bandwidth selector, and of the polynomial degree. The prediction comparisons significantly favor the parametric benchmark model. We briefly illustrate how the results of the nonparametric estimation may reveal information on the relevance of regressors and/or their functional impact, possibly allowing for further improvement of the currently preferred model. Along this line it was pointed out that kernels for discrete data differ with respect to their capabilities for smoothing out irrelevant regressors.

Nonparametric methods provide valuable insights during all modeling steps. For regressions with mixed data like the one underlying the current study, practical application of nonparametric methods only recently became available due to the collected works of Li, Racine, Hall, and others. Smoothing discrete and continuous variables with generalized product kernels allows one to overcome the potential problem of sparsely populated cells in the case of several discrete regressors, leading to a dramatic extension of applicability of nonparametric methods in applied econometrics.

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